Does Domain Fine-Tuning Change the Cascade?

Project Status Report 1 — Pipeline, Feasibility Pilot and Full Baselines

Course	CSE-645 Advanced Artificial Intelligence, Fall 2026
Project track	Project 2 — Domain-Specific Fine-Tuning of an LLM
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1. Summary

The data pipeline, evaluation harness, and frontier response cache are built and validated. All splits are deduplicated and frozen. The week-1 feasibility pilot passed its go/no-go check, and full baselines now exist for every evaluation item (5,409 items across MedMCQA calibration, MedMCQA test, and MedQA). They cover the unmodified Qwen3-8B in bf16 and in NF4, and the frontier model, Claude Opus 5.5.

Three findings shape the rest of the study. **(i) There is ample escalation headroom.** On the MedMCQA test set the frontier answers correctly on 34–37% of items that the small model gets wrong, and an oracle cascade would beat the frontier on its own (0.925 vs. 0.890). **(ii) The base model is already severely overconfident.** Mean confidence is 0.92 against 56% accuracy, and ECE is about 0.35. H1 therefore has to be framed as measuring the *direction* of change. **(iii) Both confidence signals rank items only moderately well** (AUROC ≈ 0.73), leaving substantial room for fine-tuning and recalibration to move the cascade. QLoRA training code is complete and under a smoke test; fine-tuning seed 1 is the next run.

2. Work completed

Item	Status	Detail
Repository and pipeline	**Done**	Private Git repository. Each model run writes one table with a row per item, and all analysis works offline from those tables. 38 unit tests pass.
Checkpoint pinned	**Done**	`Qwen/Qwen3-8B` at revision `b968826d`; thinking disabled; letter logits read immediately after the empty reasoning block.
Datasets pinned; splits frozen	**Done**	MedMCQA `91c6572c`; MedQA-USMLE 4-option `0fb93dd2`. Item-ID lists and hashes are committed (§3).

Letter-logit harness	**Validated**	The letters A–D are single tokens. A–D hold ≥ 96% of the next-token probability on every item (median 1.000), and A–D is the model's top prediction on 100% of items.
Frontier model pinned	**Done**	`claude-opus-5-5`, accessed 19 Oct 2026; every response cached and committed.
Feasibility pilot (n = 200)	**Passed**	Go decision (§4).
Base runs, bf16 and NF4	**Done**	Calibration, test, and MedQA splits, each item shown in its original option order plus nine fixed permutations.
Frontier full pass	**Done**	5,409 items, about $15 in total. The parser was rewritten so that it never reads the article "A" in prose as an answer.
QLoRA training code	**Smoke test**	Loss is computed on the answer tokens only, at the same position evaluation reads. Settings follow proposal §5.2.
Contamination tagging	**Pending**	Moved from weeks 2–3 to week 6 (§7).

3. Data and splits

Items were deduplicated on normalized question text plus the sorted set of answer options, and the split sizes below came out of that. Copies of a question with conflicting answer keys were dropped entirely. Items with an empty or repeated option were removed as invalid.

Split	Source	Proposal	Frozen	Note
Fine-tuning	MedMCQA train	30,000	30,000	Seeded sample from 163,221 unique valid items
Fine-tuning dev	MedMCQA train	1,000	1,000	Disjoint from fine-tuning
Calibration	MedMCQA validation	1,500	1,500	Stratified by subject
Test (primary)	MedMCQA validation	≈ 2,683	2,637	32 invalid and 14 duplicate items removed
Test (secondary)	MedQA-USMLE test	1,273	1,272	1 item with duplicate options removed

- **The training data was heavily duplicated.** 17,734 redundant copies and 851 items whose copies disagreed on the answer were removed. Only 3 training items overlapped an evaluation item.
- **The evaluation set is skewed toward one subject.** Dental makes up 31% of evaluation items, so results will also be reported per subject. A third of items (1,354) are of MedMCQA's "multi" type; each still has a single correct answer.

4. Feasibility pilot

Both models answered 200 calibration items. All the go/no-go criteria were met by a wide margin.

Measure	Value (95% bootstrap CI)
Small model accuracy (bf16)	0.555 [0.480, 0.625]
Frontier accuracy	0.900 [0.860, 0.940]
Escalation headroom (small wrong, frontier right)	0.385 [0.320, 0.450]
Both wrong / small right but frontier wrong	0.060 / 0.040
Oracle cascade	0.940 at a 38.5% call rate

5. Full baseline results

Accuracy	Calibration	Test	MedQA
Qwen3-8B, bf16	0.572	0.585	0.637
Qwen3-8B, NF4	0.558	0.562	0.622
Claude Opus 5.5 (frontier)	0.889	0.890	0.969

MedMCQA test, n = 2,637	bf16	NF4
Escalation headroom	0.340	0.365
AUROC: option probability / permutation agreement	0.739 / 0.716	0.729 / 0.725
AURC (excess AURC), option probability	0.228 (0.126)	0.252 (0.138)
ECE (15 equal-mass bins); mean confidence	0.346; 0.930	0.357; 0.918
Budget regime at a 30% call rate: probability / permutation	0.729 / 0.735	0.714 / 0.728
Oracle cascade	0.925 at 34.0%	0.926 at 36.5%

Accuracy CIs on the test set are about ±1.8 points. Budget-regime accuracies are expected values with ties broken at random. At a 30% call rate on the NF4 model, escalating random items would give about 0.66 and perfect routing about 0.86.

- **Quantization alone matters.** NF4 lowers accuracy by 1.5–2.3 points, raises ECE by 0.011, and lowers AUROC by about 0.01. All before/after-fine-tuning comparisons will therefore be made against the NF4 base, and the bf16–NF4 difference will be reported on its own.
- **The two signals rank items differently.** By AUROC, the option-probability signal ranks the whole set slightly better. At call rates of 10–50%, which depend only on the least-confident items, permutation agreement gives the better cascade. This complementarity is what the two-feature logistic combiner can exploit.

- **Frontier answers were clean.** Every response parsed except two MedQA items the frontier refused; those are scored as incorrect (2 of 1,272).

6. Implications for the hypotheses

- **H1 (fine-tuning raises ECE).** The base model's confidence is already near 1.0 on most items. If fine-tuning raises accuracy while confidence stays saturated, ECE will *fall*. The proposal's risk table anticipated this; the result is reportable either way, and the threshold regime fails for the base model already.
- **H2 (permutation agreement loses discrimination).** The baseline is now fixed: AUROC 0.725 for permutation agreement and 0.729 for option probability (NF4). Permutation agreement is the stronger signal in the budget regime at baseline.
- **H3 (cascade target).** The example target, 95% of frontier accuracy (0.846) at a 30% call rate, is far from the base model's 0.714–0.735. Closing that gap is the job of fine-tuning and recalibration.
- **H4 (recalibration).** An ECE of about 0.35 leaves a large effect for temperature scaling to repair in the threshold regime. Because the two signals rank items differently, the logistic combiner has room to improve the budget regime.
- **Contamination (§9.1 of the proposal).** On MedQA the frontier reaches 0.969 while writing about 50 output tokens per item, against about 110 on MedMCQA. Both models also score higher on MedQA than on MedMCQA. Both patterns fit MedQA's known overlap with web data. MedQA remains a consistency check only.

7. Deviations from the proposal

Change	Reason
Deduplication key includes the sorted answer options	Matching on the question alone would merge generic stems ("All of the following EXCEPT") that come with different options. Matching on question plus sorted options still catches copies whose options appear in a different order.
Test sizes 2,637 and 1,272	Invalid and duplicate items were removed (§3).
NF4 base model added as a control	QLoRA trains on an NF4 base, so comparing against bf16 would mix the effect of quantization into the effect of fine-tuning.
Frontier reasons briefly before answering	Opus 4.5 always reasons before answering, so its output was capped at 4,000 tokens. It used about 110 output tokens per item, all logged for the cost model.
Training with `transformers.Trainer` rather than `trl`	The loss masking is explicit and unit-tested, and the code does not depend on `trl`'s frequently changing API. The hyperparameters are unchanged.
Contamination tagging moved to week 6	Fine-tuning comes first because it is on the critical path; tagging adds no model runs.

8. Compute and cost to date

- **Frontier API:** about $15 for 5,409 items (about 1.0 M input and 0.52 M output tokens), within the proposal's $10–60 range. No further frontier calls are needed: escalation is simulated offline from the cache.
- **GPU:** under 1 H100-hour so far. A full evaluation pass (5,409 items × 10 option orders = 54,090 forward passes) takes about 11 minutes of scoring per model; peak memory is 17 GB in bf16 and 8 GB in NF4.
- **Environment issues, all resolved:** PyTorch is pinned to its CUDA 12.8 build (the LARCC drivers support CUDA up to 12.9), the environment is reproducible from `requirements.lock`, and the setup steps are written down for collaborators.

9. Next steps

Dates	Phase	Work
19–20 Oct	Fine-tuning	Review the smoke test, then launch seed 1 (about 1,900 steps, an estimated 1–2 H100-hours).
20–22 Oct	Fine-tuning	Evaluate seed 1 in NF4 on all three splits; first H1 and H2 readout against the NF4 base. Launch seeds 2–3.
22–31 Oct	Evaluation	Recalibration module (temperature scaling and the logistic combiner, fit on calibration only). Cascade analysis covering the budget regime and the three threshold variants: face value, threshold reuse, and threshold transfer. Bootstrap CIs and across-seed variation.
2–8 Nov	Evaluation	Contamination tagging with infini-gram and stratified results; MedQA consistency check; per-subject results.
9–22 Nov	Analysis	Error analysis (including image-dependent and malformed items), figures, publishing the adapters to the Hugging Face Hub.
23 Nov – 5 Dec	Writing	Paper draft, repository cleanup, live cascade demo, final presentation.

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